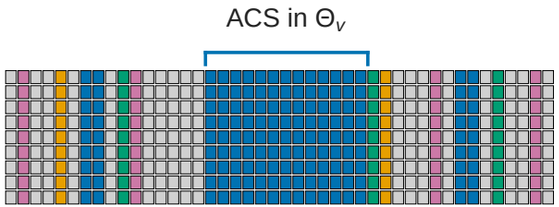
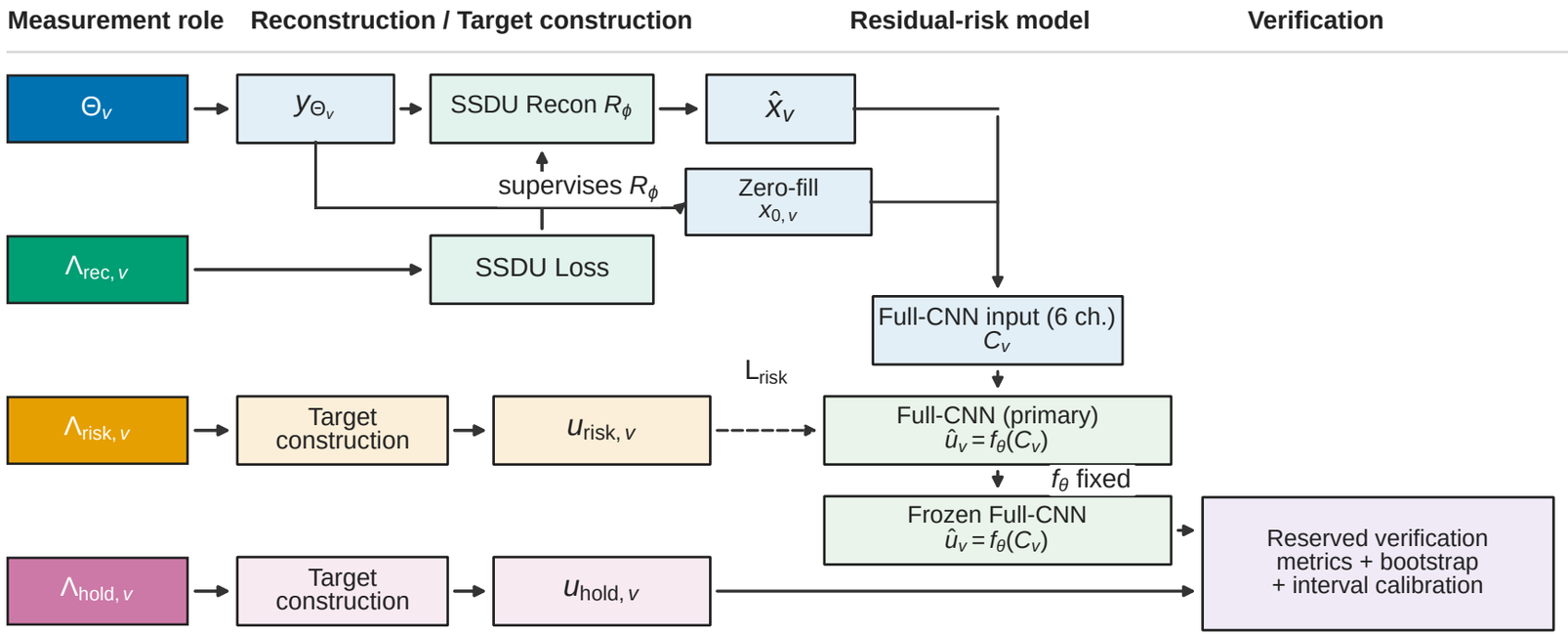
(a) Four-way acquired k-space partition

$$\Omega_v^{acq} = \Theta_v \cup \Lambda_{rec,v} \cup \Lambda_{risk,v} \cup \Lambda_{hold,v}$$



- Θ_v — reconstruction input
- $\Lambda_{rec,v}$ — SSDU reconstruction loss
- $\Lambda_{risk,v}$ — residual-risk learning
- $\Lambda_{hold,v}$ — reserved verification
- unsampled *k*-space

Role assignment is fixed per acquired phase-encode line.
ACS lines are always assigned to Θ_v.

(b) Role-separated verification workflow